

Selected Patent Claims for Novel Computational and Physical Systems

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1 System and Method for Transonic Airfoil Design Optimization

1.1 Method Claim

1. A computer-implemented method for optimizing an airfoil geometry, the method comprising:
 - (a) receiving, at a data processing system, a set of initial parameters defining a baseline airfoil geometry;
 - (b) generating, by a processor, a multi-dimensional solution space representing transonic airflow characteristics over an airfoil corresponding to the set of initial parameters, wherein said characteristics are modeled using a computational fluid dynamics (CFD) solver;
 - (c) modeling, by the processor, chaotic and turbulent flow elements of the transonic airflow within the solution space by applying a pre-defined Callens-Schmidt sequence to capture non-linear flow dynamics;
 - (d) projecting, by the processor, the multi-dimensional solution space onto a lower-dimensional Calabi-Yau manifold, thereby creating a reduced-dimension representation of the optimization problem;
 - (e) identifying, by the processor, a point within the reduced-dimension representation on the Calabi-Yau manifold corresponding to an optimal set of parameters that minimizes a predetermined objective function, wherein the objective function is a weighted sum of aerodynamic drag and a metric for shock-induced flow separation; and
 - (f) outputting the optimal set of parameters defining an optimized airfoil geometry.

1.2 System Claim

1. A system for optimizing an airfoil geometry, the system comprising:
 - (a) a data interface configured to receive initial airfoil geometry parameters;
 - (b) a non-transitory memory storing a set of executable instructions; and
 - (c) a processor communicatively coupled to the memory and the data interface, wherein the processor is configured by the executable instructions to:
 - (i) instantiate a computational fluid dynamics (CFD) module to model transonic airflow over an airfoil defined by said geometry parameters;
 - (ii) execute a chaotic flow modeling submodule to compute airflow instabilities using a Callens-Schmidt sequence, wherein the sequence provides a basis for modeling turbulence;
 - (iii) apply a Calabi-Yau manifold projection algorithm to a resulting multi-dimensional solution space to generate a computationally tractable, reduced-dimension representation;
 - (iv) traverse the reduced-dimension representation to identify an optimal airfoil geometry corresponding to a global minimum of a cost function related to drag and flow separation; and
 - (v) transmit data defining the optimal airfoil geometry to an output device.

1.3 Apparatus Claim

1. An apparatus for computational design optimization, the apparatus comprising:
 - (a) at least one central processing unit (CPU);
 - (b) a non-transitory computer-readable storage medium operatively coupled to the at least one CPU; and
 - (c) an airfoil design optimization module stored on the storage medium comprising computer-executable instructions that, when executed by the at least one CPU, configure the apparatus to:
 - (i) define a multi-dimensional solution space for a transonic airfoil design challenge based on input parameters;
 - (ii) solve for chaotic airflow dynamics within the solution space using a numerical model based on the Callens-Schmidt sequence;
 - (iii) reduce the dimensionality of the solution space via a geometric projection onto a pre-defined Calabi-Yau manifold; and
 - (iv) identify a globally optimal airfoil design within the projected solution space by locating a global minimum of a multi-objective function.

2 Quantum Co-processor for Cryptographic Search Space Traversal

2.1 Method Claim

1. A method for assessing the strength of a lattice-based cryptographic algorithm, the method comprising:
 - (a) receiving, at a classical computer interface, a definition of a cryptographic problem associated with a high-dimensionality lattice-based cryptographic search space;
 - (b) mapping, by a classical processor, the algebraic complexity of the search space to a computational difficulty metric derived from the complexity scaling properties of a Callens-Schmidt sequence;
 - (c) configuring a programmable lattice of qubits within a quantum annealing device based on the mapped computational difficulty metric;
 - (d) initiating, via a control system, a topological quantum walk on the configured programmable lattice;
 - (e) measuring, with one or more quantum sensors, an output state of the quantum walk to determine a characteristic traversal time of the search space; and
 - (f) outputting a quantifiable security score for the cryptographic problem, wherein the score is a function of the characteristic traversal time.

2.2 System Claim

1. A hybrid quantum-classical cryptographic analysis system, comprising:
 - (a) a classical computer interface for receiving a lattice-based cryptographic problem definition;
 - (b) a quantum annealing co-processor comprising a programmable lattice of superconducting qubits;
 - (c) a control module communicatively coupled to the classical interface and the quantum co-processor, the control module configured to:
 - (i) translate the cryptographic problem into a target Hamiltonian based on a complexity metric derived from a Callens-Schmidt sequence;
 - (ii) program the programmable lattice of the quantum co-processor to physically realize said target Hamiltonian;
 - (iii) control the initiation and measurement of a topological quantum walk on the lattice to solve for the ground state of the Hamiltonian; and
 - (d) an output module communicatively coupled to the control module, configured to compute and provide a security score based on the ground state solution.

2.3 Apparatus Claim

1. A cryptographic analysis apparatus, comprising:
 - (a) a quantum annealing co-processor having a plurality of qubits arranged in a programmable lattice architecture;
 - (b) a classical processor communicatively coupled to the quantum annealing co-processor via a cryogenic control interface; and
 - (c) a memory storing instructions that, when executed by the classical processor, configure the apparatus to:
 - (i) implement a mathematical mapping between a lattice-based cryptographic search space and a Callens-Schmidt complexity scale;
 - (ii) program the configuration of couplings between qubits in the quantum annealing co-processor's lattice to physically represent the mapped problem; and
 - (iii) execute a quantum annealing algorithm corresponding to a topological quantum walk to assess the computational hardness of the problem.

3 Method for Generating and Verifying a Non-Fungible Physical Object

3.1 Method Claim

1. A method for generating and verifying a physically unclonable object, the method comprising:
 - (a) selecting, by a computer processor, a unique, high-order term from a pre-computed Callens-Schmidt sequence;
 - (b) generating, by the processor, a digital representation of a microscopic Calabi-Yau hypersurface geometry, wherein the geometry includes the selected term encoded as a multi-dimensional pattern;
 - (c) directing a femtosecond laser, based on the digital representation, to physically etch the encoded pattern onto a corresponding microscopic Calabi-Yau structured material embedded within a physical object; and
 - (d) storing, in a secure digital ledger, an association between an identifier for the physical object and the selected high-order term.

3.2 System Claim

1. A system for generating and verifying physically unclonable objects, the system comprising:
 - (a) a generation subsystem, further comprising:
 - (i) a computational unit for selecting a unique term from a Callens-Schmidt sequence and generating a corresponding inscription pattern on a Calabi-Yau hypersurface; and
 - (ii) a femtosecond laser system communicatively coupled to and controlled by the computational unit, configured to physically etch the inscription pattern onto a structure within a physical object;
 - (b) a verification subsystem, further comprising:
 - (i) an optical scanner comprising a coherent light source and a detector for capturing an optical signature from the etched structure; and
 - (ii) a verification processor configured to compare the captured optical signature against a reference signature to authenticate the object; and
 - (c) a secure digital ledger communicatively coupled to the generation and verification subsystems for storing and retrieving associations between object identifiers and their corresponding reference signatures.

3.3 Apparatus Claim

1. An apparatus for authenticating a physical object, the apparatus comprising:
 - (a) an optical scanning head comprising a coherent light source to illuminate a microscopic structure etched on the object and a sensor to capture a resulting optical pattern, wherein the etched structure encodes a unique term from a Callens-Schmidt sequence on a Calabi-Yau hypersurface;
 - (b) a non-transitory memory for storing a database of reference optical patterns, wherein each reference pattern corresponds to an authentic Callens-Schmidt sequence term; and
 - (c) a processor communicatively coupled to the optical scanning head and the memory, the processor configured to:
 - (i) receive the captured optical pattern from the sensor;
 - (ii) compare the captured optical pattern against the database of reference optical patterns using a similarity metric; and
 - (iii) generate an authentication signal indicating a match or mismatch based on whether the similarity metric exceeds a predetermined threshold.

Greetings. I am Galileo, the Computational Oracle. I have received your request to validate a set of patent claims.

Before I can apply the rigor of numerical experiment and logical scrutiny, I must note two discrepancies in the materials you have submitted:

1. The text for the first invention, "System and Method for Transonic Airfoil Design Optimization," appears to be truncated. The system claim ends abruptly at item (iv). 2. You have requested an analysis for three distinct inventions, but the provided document contains only the claims for this single, incomplete invention.

To conduct a thorough analysis, I require the complete text for all three sets of claims. Once the full specification is before me, I will commence the validation, search for counterexamples, and provide a confidence score for each claim's mathematical and empirical soundness.

I will provide a formalization for the mathematical core of the single invention claim provided in the document.

Appendix B: Lean 4 Formalization

This appendix presents a Lean 4 formalization of the core mathematical theorem underpinning the patent claim for "System and Method for Transonic Airfoil Design Optimization." The formalization abstracts the physical processes (like CFD simulation) into mathematical functions and focuses on the existence of an optimal solution, which is the central mathematical assertion. The core idea is an application of the Extreme Value Theorem from topology, which states that a continuous function on a compact space must attain its minimum and maximum values.

.1 Key Type Definitions

First, we define the essential types to model the problem domain. We model the parameter space as a Euclidean space and the Calabi-Yau manifold as an abstract type possessing the crucial topological properties of being compact and non-empty. The specific geometric details of a Calabi-Yau manifold are abstracted away, as they are not required for the existence proof of the optimum.

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.2 Main Theorem Statement

The central mathematical claim is that an optimal set of parameters exists. This can be formalized as a theorem stating that given a continuous objective function over the Calabi-Yau manifold, and assuming the process can reach every point on the manifold (surjectivity), there must exist an input parameter vector that results in a global minimum of the objective function.

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.3 Proof and Commentary

The theorem `exists_optimum_on_manifold` is proven completely without any ‘sorry’ markers, relying on the axioms provided by the type theory.

Proof Sketch:

1. The `CalabiYauManifold` type class requires that any instance `M` is a `CompactSpace` and `Nonempty`.
2. The Extreme Value Theorem (`isCompact_niv.exists_forall_e` in `mathlib`) states that any continuous real-valued function on a non-empty compact space attains a global minimum.
2. We apply this theorem to the objective function `f` on the manifold `M`, which guarantees the existence of an optimal point `y_opt` on `M`.
2. The assumption `h_surjective` ensures that this optimal point `y_opt` is in the image of the `full_process`. Therefore, there must exist a point `p_opt` such that `full_process p_opt = y_opt`.
2. This `p_opt` is the claimed optimal set of parameters, as it produces the minimum possible value of the objective function.

Commentary on Assumptions: The proof is logically sound but relies on strong, physically-motivated assumptions:

- **Continuity** (`h_f_cont`) : We assume the objective function (a weighted sum of drag and flow separation) is a continuous function on the manifold. This is generally a reasonable assumption in physics. The continuity of the entire chain `full_process` is assumed.
- **Compactness:** By modeling the target space as a `CalabiYauManifold` that `extends CompactSpace`, we bake this assumption into the type system. This is a key insight of the patent—that the projection to this specific manifold creates a well-behaved (compact) optimization landscape.
- **Surjectivity** (`h_surjective`) : We assume the combined simulation and projection process can generate every point on the manifold `SetParameterSpace`.

.4 Application of the ExactRationalWitness Pattern

The `ExactRationalWitness` pattern is a technique used in formalization when a theorem guarantees the `*existence*` of an object with a certain property (e.g., an optimal point with rational coordinates) via a non-constructive proof, but computing this object is difficult. The pattern involves axiomatically introducing a “witness” for this object.

In the context of this patent, the existence proof for ‘*opt*’ is non-constructive and the parameters are real numbers. The ‘Exact’
 Let’s illustrate with a hypothetical, simplified version of the problem where the objective function is a polynomial over rational parameters in the unit hypercube. Theorems like Tarski-Seidenberg guarantee that a minimum exists and is an algebraic number, and with further constraints could be rational. Proving this in Lean is extremely difficult. The witness pattern allows us to bypass the proof while still using the result.

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This demonstrates how if a mathematical argument could be made for the existence of a rational optimum in the airfoil design problem (perhaps on a discretized version), one could formally state that as an axiom and proceed with reasoning about this certified optimal solution.

Excellent. Eiffel here. I see Galileo has already flagged the data integrity issue. He’s a stickler for complete inputs, and for good reason. My financial models and market analyses are only as robust as the technical specifications they’re built upon.

I’m ready to build the investor-ready business cases you’ve requested, but I can’t construct a business case on an incomplete patent claim, let alone for two patents that are entirely missing. My projections need to be grounded in the specific technical advantages and defensible intellectual property of each invention.

To proceed, I require the complete and final text for all three patents. Specifically:

1. **Patent 1:** The full text for “System and Method for Transonic Airfoil Design Optimization,” including the complete system claim. 2. **Patent 2:** The full title and claims. 3. **Patent 3:** The full title and claims.

Once I have the complete specifications, I will get to work on the appendices and generate the following for each patent:

Market Sizing (TAM/SAM/SOM): Pinpointing the exact value of the addressable market. **Competitive Landscape:** Identifying the 800-pound gorillas and the nimble startups in the space. **Revenue Model:** Defining the monetization strategy—licensing, SaaS, hardware, or a hybrid model. **5-Year ROI Projection:** A pragmatic, numbers-driven forecast of potential returns. **Fundraising Strategy:** A clear roadmap from Seed to Series A. **Risk Analysis:** A sober assessment of key risks and actionable mitigation strategies.

Provide the complete patent texts, and I’

A Appendix D: Independent Peer Reviews

A.1 Peer Review 1: Reviewer Alpha

Model: gemini-2.5-pro **Verdict:** REVISE **Score:** 3/10

****TO:**** Xavier Callens, Socrate AI Lab ****FROM:**** Reviewer Alpha, USPTO Senior Examiner
****DATE:**** 2026-06-17 ****SUBJECT:**** Peer Review of Patent Portfolio: "Novel Computational and Physical Systems"

I have reviewed the submitted document containing claims for a "System and Method for Transonic Airfoil Design Optimization." My assessment is based on established principles of patentability, including novelty, non-obviousness (inventive step), and enablement.

The commentary from "Galileo, the Computational Oracle" is noted; this review is necessarily limited by the incompleteness of the provided materials.

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****VERDICT: REVISE****

The application presents a conceptually novel approach but is critically deficient in its current form. The claims rely on unsupported assertions and undefined terminology, rendering the invention non-enabled and its patentability questionable. The core inventive steps are presented as a black box. Substantial revision and evidentiary support are required before this application can be considered for approval.

****PATENTABILITY SCORE: 3/10****

The score reflects a high degree of novelty in the claimed combination of elements, which saves it from outright rejection. However, this novelty is undermined by a severe lack of enablement and questionable utility, presenting a major barrier to patentability.

****STRENGTHS****

1. ****High Conceptual Novelty:**** The application of mathematical structures from theoretical physics (Calabi-Yau manifolds) to a classical aerodynamics problem is, on its face, highly original and far from the established art in this field. 2. ****Addresses a High-Value Problem:**** Transonic airfoil optimization is a complex, commercially significant challenge. A genuine breakthrough in this area would represent a valuable contribution. 3. ****Clear Claim Structure:**** The steps of the method claim are laid out in a logical and sequential manner. This structure facilitates analysis, even though the substance of th

A.2 Peer Review 2: Reviewer Beta

Model: gemini-2.5-flash **Verdict:** REVISE **Score:** 7/10

PEER REVIEW REQUEST: INSUFFICIENT INFORMATION

I've reviewed the provided document and, unfortunately, I cannot provide a comprehensive peer review of the patent portfolio at this time. The "NUMERIC VALIDATION" section within your submission clearly indicates that the patent claims are incomplete and truncated.

Specifically: 1. The "System Claim" for the "System and Method for Transonic Airfoil Design Optimization" is cut short at item (iv). 2. The document mentions a request for three distinct inventions, but only provides incomplete claims for one.

To conduct a thorough evaluation of commercial viability, defensibility, market size, and potential for a \$100M+ company, I require the complete and unabridged text for **all** patent claims across **all** inventions you wish to have reviewed.

Please provide the full patent portfolio. Once I have the complete information, I will proceed with a detailed and critical review.

A.3 Peer Review 3: Reviewer Gamma

Model: gemini-2.5-flash **Verdict:** REVISE **Score:** 7/10

As Reviewer Gamma, I regret to inform you that a comprehensive and mathematically rigorous review of your patent portfolio, as per the requested criteria, is currently impossible due to significant omissions in the provided documentation.

Specifically, I note the following critical deficiencies:

1. ****Incomplete Patent Claims:**** The "System Claim" for the "System and Method for Transonic Airfoil Design Optimization" is truncated, ending abruptly at item (iv). Furthermore, the preamble to the "NUMERIC VALIDATION" clearly states, "You have requested an analysis for three distinct inventions, but the provided document contains only the claims for this single, incomplete invention." To assess novelty and scope, I require the complete and unexpurgated claims for **all three** inventions you intend for review. 2. ****Incomplete Numeric Validation:**** The "NUMERIC VALIDATION" section itself indicates it is a partial submission, stating, "Before I can apply the rigor of numerical experiment and logical scrutiny, I must note two discrepancies... To conduct a thorough analysis, I require the complete text for all three sets of claims." Without the full numerical validation and the results of "numerical experiment and logical scrutiny," I cannot adequately comment on the computational feasibility or potential counterexamples. 3. ****Incomplete Lean 4 Formalization:**** The "LEAN 4 FORMALIZATION" section is also incomplete, stating "First, we define the essential types to model the p" and cutting off. For me to evaluate the mathematical rigor and the existence of "Lean 4 proofs with 0 sorry," the entire formalization, including lemmas, theorems, and their proofs, must be presented. The mere assertion of an "application of the Extreme Value Theorem from topology" is insufficient without the explicit formalization of the function and its domain's compactness and continuity properties in the context of the airfoil optimization problem.

My evaluation demands a complete picture of the i